CSE 304 – Web Application Development
Summary

UNIT 1

1. **Who invented the World Wide Web?** Tim Berners-Lee, a British scientist.
2. **When and where was the WWW invented?** In 1989 while Berners-Lee was working at CERN (European Council for Nuclear Research).
3. **What was the basic idea behind the Web?** To merge computer technologies, data networks, and hypertext into a global information system.
4. **Who co-authored the formalized management proposal for the Web in 1990?** Robert Cailliau, a Belgian systems engineer.
5. **What was the primary goal for scientists using the early WWW?** To browse each other's papers and documents on the Internet.
6. **When was the first proposal for the Web written?** March 1989.
7. **What did the 1990 proposal define?** It outlined principal concepts and defined important terms behind the Web.
8. **What was the first "hypertext project" called in the early documentation?** WorldWideWeb.
9. **By what time was the first Web server and browser operational?** By the end of 1990.
10. **What computer was used to develop the first Web server code?** A NeXT computer.
11. **What are the three core protocols/languages developed for Web applications?** HTML, URLs, and HTTP.
12. **What is the purpose of HTML?** For structuring web pages.
13. **What does URL stand for and what is its function?** Uniform Resource Locator; used for addressing resources on the web.
14. **What is the role of HTTP?** Transferring web data.
15. **What was the first Web address?** info.cern.ch.
16. **What was the specific address of the first Web page?** http://info.cern.ch/hypertext/WWW/TheProject.html.
17. **What warning label was placed on the first Web server?** "This machine is a server. DO NOT POWER IT DOWN!!".
18. **What unique feature did the first browser have that many modern ones lack?** The ability to modify pages directly from inside the browser (Web editing).
19. **What information did early web pages at CERN provide?** CERN phone books and guides for using central computers.

20. **How was information found on the early Web before search engines?** Through keyword-based search facilities.
21. **Who wrote the first 'line-mode' browser?** Nicola Pellow, during her student placement at CERN.
22. **Why was the 'line-mode' browser significant?** It could run on any system, whereas the original browser was limited to NeXT machines.
23. **When was the WWW software first released to the public newsgroups?** August 1991.
24. **Where was the first Web server in the US located?** The Stanford Linear Accelerator Center (SLAC) in California.
25. **Who brought the first US Web server online?** Paul Kunz and Louise Addis.
26. **Which browser, released in 1993, had a massive impact on the spread of the Web?** NCSA Mosaic.
27. **What made the Mosaic browser different from previous versions?** It offered a friendly window-based interaction and ran on PCs and Macintoshes.
28. **Name three other early browsers mentioned in the text.** MIDAS, Viola, and Erwise.
29. **When did CERN make the WWW source code royalty-free?** April 30, 1993.
30. **What percentage of internet traffic did the WWW account for in late 1993?** 1%.
31. **Which year was known as the "Year of the Web"?** 1994.
32. **What event was hailed as the "Woodstock of the Web"?** The First International World Wide Web conference at CERN in May 1994.
33. **How many servers and users were there by the end of 1994?** 10,000 servers and 10 million users.
34. **What does W3C stand for?** World Wide Web Consortium.
35. **Why was the W3C founded?** To ensure the web remained an open standard and not a proprietary system.
36. **Which organization took over the role of European W3C host from INRIA in 2003?** ERCIM.
37. **What was the first European host for W3C after CERN?** INRIA (French National Institute for Research in Computer Science and Controls).
38. **Which Asian university became a W3C host in 1996?** Keio University in Japan.
39. **What was the goal of the "WebCore" project?** To form an

international consortium in collaboration with MIT.

40. **How many member organizations did the W3C have by September 2018?** More than 400.
41. **What is the primary characteristic of a "Website"?** It delivers content from static files (text and multimedia).
42. **What is a "Web Application"?** An interactive program running on a browser that presents dynamically tailored content.
43. **What role does the browser play in the website model?** It acts as a web client or user agent that sends requests for resources.
44. **What are "web documents"?** Resources such as HTML pages, JSON, or PDF files.
45. **What does a Web Application use on the server side to manage data?** A Content Management System and databases.
46. **What are the fundamental protocols/languages for Web application design?** HTTP and HTML.
47. **Name three older Internet protocols mentioned that Web architects should know.** Telnet, FTP, and SMTP.
48. **What technologies are associated with the "Server side" of a web application?** Web servers, file systems, databases (MySQL, PostgreSQL), and business logic (PHP, Java).
49. **What is meant by the "Client side"?** The interface where users see and interact with HTML, CSS, and JavaScript.
50. **What is required of an "ideal" Web application architect?** They must be a "jack of all trades," understanding protocols, languages, databases, graphic design, and information architecture.

UNIT 2

1. **What is the foundation for Web protocols?** The TCP/IP protocol suite.
2. **When were the original Internet protocols (TCP/IP) developed?** In the 1980s.
3. **TCP/IP was an outgrowth of work done for which organization?** The United States Defense Department.
4. **What was the name of the network designed for the Defense Department?** ARPANET.
5. **What does ARPA stand for?** Advanced Research Projects Agency.

6. **What were the design goals for the networking architecture funded in the 1970s?** To be open, common, distributed, and decentralized.
7. **What was the primary flaw of centralized networks from a military perspective?** Destroying the single point of control would lose all communication.
8. **What is a "host-centric" networking architecture?** A model where devices are attached to a centralized system that coordinates all traffic.
9. **In a host-centric model, how did a terminal send text to a printer?** Data was sent to the central host first, which then parsed and sent it to the printer.
10. **How does TCP/IP treat each network device?** As a fully functional, self-aware network end-point.
11. **Can TCP/IP devices communicate directly?** Yes, without having to talk to a central host first.
12. **Why is the top-down model unsuitable for millions of devices?** Because it would not work with widely distributed devices on a national scale.
13. **How does the decentralized design provide reliability?** The network continues to function even if part of it is damaged.
14. **What was a major issue with early proprietary network architectures?** They limited interoperability between different systems.
15. **What was the foremost design goal for the new architecture funded by the Defense Department?** Establishing a decentralized, distributed network topology.
16. **Where does network 'intelligence' reside in a distributed topology?** It is distributed among many systems throughout the network.
17. **What technology allows a message to be split and take different routes?** Packet-switching technology.
18. **Can packets arriving in mixed-up order still be understood?** Yes, they are reassembled by the intended recipient.
19. **What group was formed to examine connecting heterogeneous networks?** The Internet Working Group (INWG).
20. **What did the INWG eventually evolve into?** Bodies like the IAB, IANA, IETF, and IESG.
21. **What does IAB stand for?** Internet Architecture Board (formerly Internet Activities Board).
22. **What does IANA stand for?** Internet Assigned Numbers Authority.

23. **What does IETF stand for?** Internet Engineering Task Force.
24. **What are RFCs?** Requests for Comments; proposals for new or updated protocols.
25. **What happens once an RFC is approved?** It is treated as the standard documentation for that protocol.
26. **What two protocols give the TCP/IP suite its name?** Transmission Control Protocol (TCP) and Internet Protocol (IP).
27. **Is TCP/IP a single monolithic protocol?** No, it is a modular family of protocols with segmented functions.
28. **What are the four layers of the TCP/IP stack?** Network Interface, Internet, Transport, and Application .
29. **What is another term for these layered implementations?** A protocol stack.
30. **Which layer is responsible for the lowest level of data transmission?** The Network Interface layer.
31. **What are the functions of the Internet layer?** Inter-system communications, message routing, and validity checking.
32. **Which two specific protocols operate at the Internet layer?** IP and ICMP.
33. **What does ICMP stand for?** Internet Control Message Protocol.
34. **What is the purpose of ICMP?** Handling the transmission of control and error messages.
35. **Which service uses ICMP to check if devices are online?** Ping.
36. **What does the Transport layer provide?** Message transport services between applications on remote systems.
37. **What are the characteristics of TCP?** Reliable and connection-oriented.
38. **What protocol is used by services that trade reliability for performance?** UDP (User Datagram Protocol).
39. **Why do streaming services prefer UDP?** To get faster performance by sacrificing a few lost packets.
40. **What is the highest level within the TCP/IP stack?** The Application layer.
41. **What does OSI stand for?** Open Systems Interconnection.
42. **Who developed the OSI model?** The International Standards Organization (ISO).
43. **How many layers are in the OSI model?** Seven.
44. **List the seven OSI layers.** Physical, Data Link, Network, Transport,

Session, Presentation, and Application.

45. **What is the Physical layer concerned with?** Physical wiring, cables, connectors, and jacks .
46. **What is the responsibility of the Data Link layer?** Defining how information is transmitted across the physical layer and dealing with errors.
47. **What is the purpose of the OSI Network layer?** Identifying system addresses and the actual transmission of data.
48. **What does the OSI Transport layer provide?** Reliability services for basic transmission.
49. **What is the function of the Session layer?** Establishing and destroying connections between systems or applications.
50. **What is an example of a service provided by the Presentation layer?** Data compression.
51. **Does the Application layer include the part of the software that displays data to users?** No, that is outside the scope of the OSI model.
52. **What application protocol does a mail client use to send messages?** SMTP.
53. **What is the NVT specification?** Network Virtual Terminal; rules used when SMTP exchanges data with TCP.
54. **Which TCP/IP layer corresponds to the OSI Network layer?** The Internet layer.
55. **Which TCP/IP layer combines OSI's Session, Presentation, and Application layers?** The Application layer.
56. **Does TCP/IP have services that correspond directly to the OSI Physical or Data Link layers?** No, it relies on the local operating system and device drivers.
57. **What is the Internet Protocol (IP)?** A set of rules allowing devices to communicate over the Internet.
58. **How does IP ensure information reaches the correct destination?** By using unique IP addresses.
59. **Is IP a connection-oriented or connectionless protocol?** Connectionless.
60. **What is an IP address?** A unique number sticker given to each device on a network.
61. **What are the two main purposes of an IP address?** Host/network recognition and location addressing.
62. **What is a "Packet"?** A parcel of data switched between origin and destination.

63. **What are the two main components of a packet?** A header and a payload.

64. **What is the "payload" in a packet?** The actionable data being transmitted.

65. **What is a "Router"?** A network device that serves as a forwarding point for data packets.

66. **What is IPv4?** Internet Protocol version 4, covering most internet communications today.

67. **What is IPv6?** The final iteration of IP updates used for identifying and routing computers.

68. **What is the "phonebook of the internet"?** Domain Name System (DNS).

69. **What does DNS do?** Translates human-readable domain names into IP addresses.

70. **What protocol maps IPs to physical MAC addresses for IPv4?** ARP.

71. **What protocol maps IPs to physical MAC addresses for IPv6?** NDP.

72. **What are the two protocols used for routing through network devices?** OSPF or BGP.

73. **What does NAT allow?** Multiple devices to share a single public IP.

74. **What are two benefits of NAT?** Conserving address space and enhancing security by hiding internal addresses.

75. **What is "Static IP configuration"?** Manually assigning a permanent, unchanging IP address to a device.

76. **What does DHCP stand for?** Dynamic Host Configuration Protocol.

77. **What is the benefit of DHCP?** It automatically assigns network details, preventing IP conflicts and simplifying management.

78. **What is the basic unit of data transmission in an IP network?** An IP packet.

79. **What information does an IP header contain?** Essential control information like source and destination addresses.

80. **What determines the best route for an IP packet?** Routers examine the IP header.

81. **What is "decapsulation"?** The process of removing headers and passing the payload to the application upon arrival.

82. **What is the size of an IP header in the provided diagram?** 24 Bytes.

83. **What is the total width of the packet rows in bits?** 32 bits (4 Bytes).

84. **What does "Time to Live" (TTL) refer to in a packet?** A field in the IP header.

85. **What is the "Header Checksum" used for?** Validity checking of the header.

86. **What is IP routing?** The procedure of directing information from source to recipient.

87. **What determines the path a data package follows?** The routing algorithm.

88. **What factors does a routing algorithm consider?** Packet size, header length, and determining the most suitable route.

89. **What is a "routing table" used for?** To decide the subsequent hop's address.

90. **Can packets of the same message travel through different routes?** Yes, they are independent packets.

91. **Name four Application layer protocols.** HTTP, FTP, Telnet, SMTP .

92. **Name two Network Interface layer protocols.** Ethernet and Token Ring.

93. **What happens to data as it travels from the top layer to the bottom?** It becomes more encapsulated with additional headers.

94. **What is the function of TCP in an operating system?** To confirm if a packet reached its destination or needs retransmission.

95. **What is the function of HTTP for a web browser?** To understand how to display a webpage.

96. **How does an email server use the TCP layer?** By dividing data into packets and providing numbers to each.

97. **Where does the IP layer send the packets in the email example?** To the destination email server.

98. **What does the recipient server's TCP layer do?** Separates data packets from the IP layer to restore original text.

99. **What is "Padding" in an IP packet?** A field used to ensure the header ends on a 32-bit boundary.

100. **What does the IP layer use to route packets to the public internet?** The operating system.

UNIT 3

1. **What does Telnet stand for?** Teletype Network.

2. **What type of application protocol is Telnet?** A client/server application protocol.

3. **What service does Telnet provide?** Access to virtual terminals of remote systems on local area networks or the Internet.

4. **Which program does the local computer use in a Telnet session?** A telnet client program.

5. **Which program do remote computers use for Telnet?** A telnet server program.
6. **Is Telnet based on TCP or UDP?** It is a TCP-based protocol.
7. **What port number does Telnet use?** Port 23.
8. **Why does Telnet use TCP instead of UDP?** To establish a reliable, connection-oriented session and ensure data integrity.
9. **Which layer of the protocol stack does Telnet operate in?** The Application layer.
10. **What is "Network Virtual Terminal" (NVT) functionality?** The ability to log in to a remote machine over the Internet using a common structure shared by different terminals.
11. **Where is the latest specification for Telnet defined?** In Internet RFC 854.
12. **What is the "local computer" in a Telnet connection?** The computer that starts the connection.
13. **What is the "remote computer"?** The computer being connected to that accepts the connection.
14. **What are the two categories of the logging process?** Local Login and Remote Login.
15. **What is "Local Login"?** When a user logs into their own local system.
16. **What is "Remote Login"?** A process where users log in to a remote site to use services available on that computer.
17. **What piece of software on the remote machine pretends characters are coming from a terminal?** A pseudo-terminal driver.
18. **What does the Telnet client do with typed characters?** It transforms them into NVT characters.
19. **How do NVT characters travel through the Internet?** Via the local TCP/IP protocol stack.
20. **What is the primary purpose of NVT?** To ensure communication compatibility between various terminals with different operating systems.
21. **What is the decimal code for the Telnet command "WILL"?** 255.
22. **What does the "WILL" command mean?** Offering to enable or accepting a request to enable a feature.
23. **What is the decimal code for "WON'T"?** 252.
24. **What does "WON'T" signify?** Rejecting a request to enable, offering to disable, or accepting a request to disable.
25. **What is the decimal code for "DO"?** 253.

26. **What does "DO" mean?** Approving a request to enable or requesting to enable.
27. **What is the decimal code for "DON'T"?** 254.
28. **What does "DON'T" signify?** Disapproving a request to enable, approving an offer to disable, or requesting to disable.
29. **What does "IAC" stand for?** Interpret As Command.
30. **What is the decimal code for IAC?** 255.
31. **What is the general syntax for a Telnet command?** IAC Command Code options.
32. **What are the two main phases of establishing a Telnet connection?** 1. TCP Connection Establishment and 2. Telnet Option Negotiation.
33. **What is another name for the TCP connection phase?** Three-Way Handshake.
34. **In the three-way handshake, what is the first segment sent by the client?** SYN.
35. **What does the server send back after receiving a SYN?** SYN-ACK.
36. **What is the final step of the handshake?** The client sends an ACK.
37. **What features are typically negotiated during Telnet option negotiation?** Echo control, Terminal type, Window size, and Binary transmission.
38. **When does the username/password prompt appear?** After the TCP connection and Telnet option negotiation are established.
39. **What is Electronic Mail?** A method of exchanging digital messages over the Internet using electronic devices.
40. **What types of content can be sent via email?** Text, files, images, videos, and documents.
41. **What is a "User Agent" (UA)?** An email client or application used to compose, read, and manage emails (e.g., Gmail).
42. **What is a "Mail Server"?** A centralized system that stores emails and manages transmission.
43. **What is the role of a "Message Transfer Agent" (MTA)?** Routing emails from the sender's server to the recipient's server.
44. **What is a "Mailbox"?** A storage location on the mail server where received emails are kept.
45. **What is a "Spool File"?** A temporary holding area for outgoing emails before they are processed.
46. **Which protocol is used by the client to send a message to the**

mail server? SMTP (Simple Mail Transfer Protocol).

47. **How does an MTA find the receiver's mail server?** Via DNS/MX records.
48. **Which protocols are used by the client to access or receive mail from a mailbox?** IMAP or POP3.
49. **Is SMTP a connectionless or connection-oriented protocol?** Connection-oriented.
50. **Which layer of the TCP/IP stack does SMTP operate in?** The Application layer.
51. **Which transport layer protocol does SMTP use?** TCP.
52. **On what port does an SMTP server usually run?** Port 25.
53. **Does SMTP handle incoming or outgoing mail?** It handles outgoing mail.
54. **What is a "push protocol" in the context of email?** A protocol like SMTP that sends messages toward the destination.
55. **What are three common ports for SMTP?** 25, 587, and 465.
56. **What does the SMTP command "HELO" do?** Initiates the conversation between the client and server.
57. **What does the "MAIL FROM" command identify?** The sender's email address.
58. **What does "RCPT TO" specify?** The recipient's email address.
59. **What does the "DATA" command signify?** The beginning of the actual message body.
60. **What does "QUIT" do in SMTP or POP3?** Terminates the connection.
61. **What does "POP" stand for?** Post Office Protocol.
62. **What is the primary use of POP?** To retrieve email for a single client by downloading it to a device.
63. **What is the current version of POP in use?** POP3.
64. **Does POP3 support offline access?** Yes, because messages are downloaded to the local device.
65. **What typically happens to messages on the server after POP3 downloads them?** They are often removed/deleted.
66. **What is the non-secure port for POP3?** Port 110.
67. **What is the secure port for POP3?** Port 995.
68. **Does POP allow multiple mailboxes on a server?** No, it allows only a single mailbox.
69. **Does POP3 provide search facilities?** No.
70. **What does the "UIDL" command do in POP3?** Provides a Unique Identifier Listing of messages to determine which are "new".

CSE 304 – Web Application Development

Summary

71. **What does "IMAP" stand for?** Internet Message Access Protocol.
72. **What is the main advantage of IMAP over POP3?** It syncs emails across multiple devices and keeps them on the server.
73. **Where are IMAP emails maintained?** On the remote server.
74. **Does IMAP support search facilities?** Yes.
75. **What is the non-secure port for IMAP?** Port 143.
76. **What is the secure port for IMAP?** Port 993.
77. **What happens when you organize folders in IMAP?** The changes are synced back to the server and all devices.
78. **Which protocol is best for users who check mail on multiple devices?** IMAP.
79. **What are message forums?** Online services for posting and reading messages on electronic bulletin boards.
80. **What was likely the earliest form of online message forums?** Electronic mailing lists.
81. **What is the difference between a mailing list and a public forum?** Mailing lists are closed systems (subscribers only), while forums are for open participation.
82. **What was FIDONET?** A cooperative network that allowed systems to share and forward messages.
83. **What is the Internet version of message forums?** Netnews.
84. **What are the main divisions (hierarchies) of Netnews?** comp, sci, soc, talk, and alt.
85. **What does "comp" stand for in Netnews?** Computing-related newsgroups.
86. **What does "alt" represent?** An unregulated hierarchy for "alternative" newsgroups.
87. **What protocol is used to transmit Netnews?** NNTP (Network News Transfer Protocol).
88. **Where is the NNTP specification defined?** In Internet RFC 977.
89. **What site (now part of Google) created a powerful search engine for newsgroups?** deja.com.
90. **What protocol enabled network-connected UNIX users to talk to each other before AOL?** The talk protocol.
91. **Why did talk servers have to run on both ends of a conversation?** Because talk was a bi-directional service.
92. **Has an open, interoperable Instant Messaging protocol been successfully established?** No, most are currently proprietary and exclusionary.

93. **What was used to share files before the Internet?** Electronic Bulletin Board Systems (BBS).
94. **Name three file transfer protocols used over telephone lines.** Kermit, Xmodem, and Zmodem.
95. **What does "FTP" stand for?** File Transfer Protocol.
96. **Where is the most recent version of the FTP specification found?** In Internet RFC 959.
97. **How many connections does FTP usually require?** Two.
98. **What are the two types of FTP connections?** 1. Control connection (for commands) and 2. Data connection (for actual data).
99. **What is "anonymous FTP"?** A service that allows open access to files without explicit user authentication.
100. **What is the standard password for anonymous FTP according to 'Netiquette'?** The user's email address

UNIT 4

1. **What is the primary subject of Module 4?** The Hypertext Transfer Protocol (HTTP).
2. **Who devised the three basic components of Web technology?** Tim Berners-Lee.
3. **What are the three basic components of the Web?** 1. A markup language (HTML), 2. A uniform notation scheme (URI/URL), and 3. A transport protocol (HTTP).
4. **What does HTML stand for?** HyperText Markup Language.
5. **What is the primary purpose of HTML?** Formatting hypertext documents and allowing cross-referencing via hyperlinks.
6. **What does URI stand for?** Uniform Resource Identifier.
7. **What does URL stand for?** Uniform Resource Locator.
8. **What was the original meaning of the 'U' in URL?** Universal.
9. **What is a URN?** Uniform Resource Name.
10. **What is the difference between a URL and a URN?** URLs embed server and file locations, while URNs use a fixed, immutable name that stays the same even if a resource is moved.
11. **Why do we still use URLs despite URNs being considered a "nicer mechanism"?** URNs failed to materialize as a globally supported web standard.
12. **How does the W3C define a URI?** As the union of URLs and URNs.
13. **Which term is more formally correct: URL or URI?** URI.

CSE 304 – Web Application Development

Summary

14. **What is the generalized notation of a URL?** scheme://host[:port][/path/][;url-params][?query-string][#anchor].
15. **What does the 'scheme' portion of a URL designate?** The underlying protocol to be used, such as http or ftp.
16. **Where is the scheme located in a URL?** Preceding the colon and two forward slashes.
17. **What can the 'host' part of a URL be?** Either a name or an IP address for the web server.
18. **What is the default port number for HTTP?** 80.
19. **When must a port number be specified in a URL?** When the server configuration uses an alternate port number other than the default.
20. **What is the 'path' in a URL?** Logically, it is the file system path from the root directory to the document.
21. **What is 'aliasing' in web servers?** Pointing to documents or services not explicitly accessible from the server's root directory.
22. **What are 'url-params' used for today?** Often for session identifiers in web servers supporting the Java Servlet API.
23. **What character precedes URL parameters?** A semi-colon (;).
24. **What does the 'query-string' contain?** Dynamic parameters, usually from user-entered variables in HTML forms.
25. **What character introduces a query string?** A question mark (?).
26. **In a query string, what separates parameters from values?** Equal signs (=).
27. **What separates multiple parameter-value pairs in a query string?** Ampersands (&).
28. **What is an 'anchor' in a URL?** A reference to a positional marker (bookmark) within a document.
29. **What character precedes an anchor?** A hash mark or pound sign (#).
30. **Is HTTP an application layer or transport layer protocol?** Application level protocol.
31. **What transport layer protocol does HTTP use?** TCP.
32. **What paradigm does the HTTP protocol use?** The request/response paradigm.
33. **What is an HTTP transaction composed of?** A single request from a client followed by a single response from the server.
34. **What does it mean that HTTP is a "stateless" protocol?** It has no explicit support for maintaining the state of a connection throughout successive commands.

35. **What other internet protocols are "stateful"?** FTP, SMTP, and POP.
36. **What mechanism is commonly used to maintain state in Web applications?** Cookies.
37. **How does HTTP/1.1 improve efficiency regarding connections?** It supports persistent connections that outlive a single request/response exchange.
38. **Does HTTP/1.1 support state?** No, it supports persistent connections for efficiency, but it remains stateless.
39. **What are HTTP servers often called?** HTTP servers or Web servers.
40. **What are Web browsers sometimes called in HTTP contexts?** HTTP clients.
41. **Did Tim Berners-Lee intend browsers to only support HTTP?** No, they were designed to enable access to a wide variety of content, including FTP and Gopher.
42. **What are HTTP proxies?** Programs that act as both servers and clients, making requests to web servers on behalf of other clients.
43. **What are three functions of a proxy?** 1. Enabling transfers across firewalls, 2. Caching messages, and 3. Filtering requests.
44. **What other protocol has a message structure similar to HTTP?** E-mail.
45. **What are the three main sections of an HTTP message?** 1. Message headers, 2. A blank line, and 3. A message body.
46. **What is the first line of an HTTP request called?** The request line.
47. **What three fields are in a request line?** 1. Method, 2. Path to resource, and 3. Version number.
48. **What is a <CR><LF>?** A carriage return and line feed, forming a blank line.
49. **What is the difference between a carriage return (\r) and a line feed (\n)?** Carriage return moves the cursor to the beginning of the line; line feed moves it to the next line.
50. **What signals the end of an HTTP request message?** A blank line following the body.
51. **Which header is required in every HTTP/1.1 request?** The Host header.
52. **What is the first line of an HTTP response called?** The status line.
53. **What three fields are in a status line?** 1. Version number, 2. Status code, and 3. Human-readable message.
54. **What is a "success code" in HTTP?** 200 OK.

55. **How does a browser handle an HTML page containing images?** It parses the page to find additional resources and sends separate HTTP requests to retrieve them.
56. **Are HTTP messages intended to be human-readable?** No, unlike e-mail, they are intended for sophisticated interactions between machines.
57. **What are the three most basic request methods in HTTP/1.1?** GET, HEAD, and POST.
58. **List five less common HTTP request methods.** PUT, DELETE, TRACE, OPTIONS, and CONNECT.
59. **Which method is used when you click a hyperlink or enter a URL?** GET.
60. **Does a GET request have a body?** No.
61. **Where does a GET request put form parameters?** In the query string appended to the URL.
62. **What is a fundamental difference between GET and POST?** POST requests have a body where form parameters are placed, while GET appends them to the URL.
63. **When might a developer use both GET and POST for the same URL?** To display a form using GET and then process that form using POST.
64. **What does a server return in response to a HEAD request?** Only the headers (metadata), with no body.
65. **Why is the HEAD method useful?** It reduces overhead and bandwidth by allowing clients to check metadata without downloading the full content.
66. **What was the historical use of HEAD requests?** To implement caching support.
67. **What is "caching"?** Storing copies of frequently accessed data in a high-speed storage layer to reduce latency.
68. **What header allows a client to check if a resource was changed since a specific date?** Last-Modified.
69. **How many categories of response messages are in HTTP/1.1?** Five.
70. **What do status codes starting with '1' represent?** Informational status codes.
71. **What is the purpose of informational status codes?** To impart information on how a request can be processed further.
72. **What does a status code of 100 mean?** Continue; the client may proceed with a partially submitted request.
73. **What do status codes starting with '2' represent?** Successful responses.

74. **What does status code 201 indicate?** The request was satisfied and a new resource was created.
75. **What do status codes starting with '3' represent?** Redirection.
76. **What action does a 3xx code require of the client?** Additional actions to satisfy the original request, normally redirecting to another URL.
77. **Where is the new location specified in a redirection response?** In the Location header.
78. **What is the difference between 301 and 302?** 301 is "Moved Permanently"; 302 is "Moved Temporarily".
79. **How do browsers typically handle 301 and 302 codes?** They react silently so users never see the redirection happen.
80. **What happens if you enter a directory URL without a terminating slash?** The server typically performs a silent 301 redirection to the URL with the slash.
81. **Why is it important for applications to include the trailing slash in links?** To avoid an extra connection and request caused by server-side redirection.
82. **What do status codes starting with '4' represent?** Client request errors.
83. **What does code 400 indicate?** Bad Request.
84. **What does code 401 indicate?** Not Authorized (authorization challenge).
85. **What does code 404 indicate?** Not Found.
86. **What does code 412 indicate?** Precondition Failed (e.g., used with If-Unmodified-Since).
87. **What does code 417 indicate?** Expectation Failed.
88. **What do status codes starting with '5' represent?** Server errors.
89. **What does code 500 indicate?** Internal Server Error.
90. **What does code 501 indicate?** Not Implemented.
91. **Which Internet RFC documents HTTP status codes?** RFC 2616.
92. **How does a browser react to a 301 Moved Permanently response to be more efficient?** It updates an internal relocation table to use the new address directly in the future.
93. **Why do servers include a message body in 301 responses?** To support older browsers that do not automatically relocate.
94. **What is the effect of omitting a trailing slash in a directory link?** It effectively doubles the number of requests sent to the server.

95. **What is MIME functionality?** Extensions to e-mail structure to support decompression, decoding, and attachments.

96. **Is the human-readable explanation of a status code required for a browser to function?** No, changing it will not cause a properly designed client to change its actions.

97. **Can proxies filter HTTP requests?** Yes.

98. **What was Tim Berners-Lee's markup language called?** HTML.

99. **Is HTTP simple or complex?** It is considered a simple protocol, though interactions can become complex in sophisticated applications.

100. **Does a HEAD request use more or less bandwidth than GET?** Less, because it does not include the content body.

UNIT 5

1. **What are HTTP headers considered in a message?** They are a form of message metadata.

2. **What can sophisticated applications achieve using headers?** They can establish sessions, set caching policies, control authentication, and implement business logic.

3. **What four categories of headers does the HTTP specification define?** General headers, request headers, response headers, and entity headers.

4. **What is a General header?** A header that applies to both request and response messages but does not describe the message body.

5. **What does the Date header specify?** The time and date the message was created.

6. **What is the function of the Connection: Close header?** It indicates if the sender intends to keep the connection open or close it.

7. **What is the Warning header used for?** It stores text for human consumption, useful for tracing problems.

8. **What do Request headers allow clients to do?** Pass additional information about themselves and the request to the server.

9. **Which header identifies the software (like a browser) making a request?** The User-Agent header.

10. **Why was the Host header introduced?** To support virtual hosting, allowing one server to service multiple domains.

11. **What information does the Referer header provide?** The URL of the page containing the link that

the user clicked to make the current request.

12. **When is the Authorization header used?** When requesting resources restricted to authorized users.
13. **What status code triggers a browser to prompt for credentials?** 401 Unauthorized.
14. **How long does a browser typically supply credentials after the initial prompt?** For all further requests during the current session within the same authorization realm.
15. **What do Response headers help a server communicate?** Additional information about the response that status codes alone cannot provide.
16. **Which header accompanies 301 and 302 status codes to specify a new URL?** The Location header.
17. **What does the WWW-Authenticate header specify?** The protected realm for which credentials must be provided.
18. **Is the Server header tied to a specific status code?** No, it is an optional header identifying the server software.
19. **What do Entity headers describe?** The message body or the target resource if no body is present.
20. **Which header specifies the MIME type of a message body?** Content-Type.
21. **What is the benefit of including a Content-Length header for a browser?** It allows the browser to display download progress as a percentage.
22. **Why is the Last-Modified header critical?** It is essential for the proper functioning of caching mechanisms.
23. **Why is content important in Web applications?** Without it, there would be nothing for users to see or interact with.
24. **Name three things a browser might do with content.** Render it as HTML, launch a helper application, or use a plug-in.
25. **Where does HTTP borrow its content typing system from?** Multipurpose Internet Mail Extensions (MIME).
26. **What was MIME originally designed for?** To help e-mail clients display non-textual content.
27. **What is the two-layer ordered encoding model for HTTP bodies?** It uses Content-Type and Content-Encoding headers.
28. **Must a body be decoded before being interpreted by its Content-Type?** Yes, it must be decoded according to the Content-Encoding header first.
29. **Name the three standard encoding methods in HTTP/1.1.** "gzip", "compress", and "deflate".

30. **Which encoding methods are equivalent to "x-gzip" and "x-compress"?** "gzip" and "compress".

31. **How does a browser handle a "Content-Encoding: gzip" header for a .doc file?** It decodes the file and invokes Microsoft Word to view it.

32. **What is the structure of a media-type in a Content-Type header?** type "/" subtype [optional parameter string].

33. **What does "Content-Type: text/html" tell a browser?** To render the body as an HTML page.

34. **In "text/plain; charset='us-ascii'", what is the parameter string?** charset='us-ascii'.

35. **What happens if a client program doesn't recognize a Content-Type parameter?** It is simply ignored.

36. **Give an example of a MIME type for XML.** "text/xml" or "application/xml".

37. **What is the MIME type for a PDF?** "application/pdf".

38. **What does a multipart message allow?** The inclusion of multiple independent entities within a single message body.

39. **How can a server use multipart messages to help a client?** It can implement primitive image animation by feeding a sequence of images.

40. **What does "multipart/x-mixed-replace" instruct a browser to do?** Render enclosed images one at a time in the same screen location.

41. **How are individual parts of a multipart message separated?** By a boundary string specified in the header.

42. **Why do server-side applications need type information for file uploads?** To separate the file content from accompanying form data like the file name.

43. **What is caching in HTTP?** Storing responses in a temporary medium to improve performance and throughput.

44. **What are the three main types of caching?** Server-side, browser-side, and proxy-side caching.

45. **Who usually decides if a cached response is appropriate?** The server or the applications running on it.

46. **What can a server do if it knows content is static?** Instruct clients to cache the response for a long period.

47. **Can a server allow caching for constantly changing content?** Yes, if it decides the client can tolerate out-of-date content for a short time.

48. **Which header enforces caching rules in HTTP/1.1?** Cache-Control.
49. **What does the public cache setting mean?** Both shared and non-shared caches can store the response without restriction.
50. **What does the private setting indicate?** The response is for a single user and should not be stored in a shared cache.
51. **Why is the private setting important for secure account data?** It prevents a proxy from sending one user's private data to a second user.
52. **What does no-cache signify?** Browsers and proxies are generally not allowed to cache the response.
53. **Can specific headers be excluded from a cache?** Yes, by listing them in the Cache-Control header.
54. **Are HTTP/1.0 agents guaranteed to obey the Cache-Control header?** No, it was introduced in HTTP/1.1.
55. **What is the purpose of the deprecated Pragma header?** It has one setting, no-cache, to prevent HTTP/1.0 agents from caching.
56. **What is the best backwards-compatible solution for disabling cache?** Including both Pragma: no-cache and Cache-Control: no-cache headers.
57. **How are credentials transmitted in basic HTTP authentication?** As a single encoded string in the Authorization header.
58. **Is basic authentication secure over an open connection?** No, because the string is encoded (not encrypted) and easily decoded.
59. **How can you tell if an application is using built-in HTTP authentication?** The browser displays its own dialog box instead of an HTML page prompt.
60. **Where do application-specific schemes prompt users?** In the main browser window as part of a rendered HTML page.
61. **What code does a server send to initiate a security challenge?** 401 Authenticate.
62. **What is the purpose of the "realm" name?** It helps users retrieve passwords and organizes resources requiring specific authorization.
63. **What encoding scheme is used for Basic authentication?** Base64.
64. **What character separates the username and password before encoding?** A colon.
65. **What happens if a server fails to verify credentials?** It may resend the 401 challenge or send a 403 Forbidden response.

CSE 304 – Web Application Development

Summary

66. **Why do servers limit security challenges?** To prevent break-ins by trial-and-error.
67. **How do most commercial applications handle sensitive financial logins?** They use proprietary schemes, often POSTing data over secure connections.
68. **What is the main difference between encoding and encryption?** Encryption requires a unique key to decode, whereas simple encoding can be reversed by anyone who knows the scheme.
69. **What should a user check before entering a password in an HTTP-based prompt?** If the URL uses https (Secure HTTP).
70. **What is a "Dependent URL"?** A URL that shares a prefix with an authorized URL up to the last slash.
71. **If /test/ is authorized, will the browser send credentials for /test/classes/?** Yes, because it is a dependent URL.
72. **What mechanism established state in HTTP/1.1?** Set-Cookie and Cookie headers.
73. **Which header does a server use to set attributes in a browser?** Set-Cookie.
74. **Which header does a browser transmit in subsequent requests?** Cookie.
75. **What attributes can be included in a Set-Cookie header?** name, value, expires, path, domain, and secure.
76. **Where do browsers store cookies that only last for the current session?** In an in-memory table.
77. **Where does Internet Explorer keep persistent cookies?** In a folder where each file represents a particular cookie.
78. **What is the rule for setting a cookie's domain?** It must be the same domain as the server or a valid suffix (e.g., .rutgers.edu).
79. **What does a domain value of .rutgers.edu mean?** The cookie applies to all hosts ending in .rutgers.edu.
80. **What is the default value for the path attribute?** The path of the URL of the request.
81. **How long does a cookie last if no expires date is specified?** Only for the duration of the current browser session.
82. **What does the secure keyword tell a browser?** To only pass the cookie over secure connections.
83. **Why is backward compatibility a challenge for HTTP evolution?** Many programs still use HTTP/1.0 and don't always handle 1.1 messages correctly.

84. **What is Virtual Hosting?** The ability to map multiple host names to a single IP address.
85. **How did HTTP/1.0 requests via proxies identify the destination server?** By including the full URL in the initial request line.
86. **Why do HTTP/1.1 requests require the Host header?** To preserve destination information even if an HTTP/1.0 proxy removes the server info from the request line.
87. **What happens if an HTTP/1.1 request is missing the Host header?** It is an illegal request format for HTTP/1.1.
88. **What was the popular caching mechanism for HTTP/1.0?** Using HEAD requests to check the Last-Modified header.
89. **What two headers does HTTP/1.1 use for more streamlined caching?** If-Modified-Since and If-Unmodified-Since.
90. **What status code is returned if a resource hasn't changed in an If-Modified-Since request?** 304 Not Modified.
91. **What status code is returned if an If-Unmodified-Since precondition fails?** 412 Precondition Failed.
92. **Why is the cost of connecting to a server considered "considerable"?** Because of the network overhead required to establish each connection.
93. **Was HTTP originally designed for persistent connections?** No, it was intended to last only for one request and response.
94. **What was the proprietary workaround for persistent connections in HTTP/1.0?** The Connection: Keep-Alive header.
95. **Why did Keep-Alive often fail in HTTP/1.0?** One intermediate proxy without support was enough to break the connection.
96. **Are HTTP/1.1 connections persistent by default?** Yes.
97. **How can an HTTP/1.1 program explicitly request to end a connection?** By using the Connection: Close header.
98. **Does Connection: Keep-Alive have any legal meaning in pure HTTP/1.1?** No, since persistence is the default.
99. **What is "pipelining" in HTTP/1.1?** It allows browsers to queue multiple requests without waiting for each response first.
100. **In what order must servers submit responses to pipelined requests?** In the order of their arrival.

UNIT 6

1. **What is a "Web site" defined as in this module?** A collection of

documents and information organized in a tree structure similar to a file system.

2. **What is the primary function of a Web server?** To enable HTTP access to a website.
3. **What does a modern Web server provide besides access to static documents?** It implements protocols to pass requests to custom software for dynamic content.
4. **What are three examples of dynamic content sources?** Search engines/databases, measuring instruments, and news feeds .
5. **What does CGI stand for?** Common Gateway Interface.
6. **Name four proprietary template languages mentioned as alternatives to CGI.** PHP, Cold Fusion, Active Server Pages (ASP), and Java Server Pages (JSP).
7. **What is the "ideal approach" for serving dynamic data?** A declarative fashion where site maintainers are not required to write custom code.
8. **What three entities communicate by exchanging HTTP messages?** Web servers, browsers, and proxies.
9. **Which module is responsible for receiving and transmitting network signals?** The Networking module.
10. **To which module is a request first passed after being received?** The Address Resolution module.
11. **What is the function of the Address Resolution module?** Analyzing and "pre-processing" the request.
12. **What three tasks are involved in pre-processing?** Virtual hosting, address mapping, and authentication .
13. **What is the role of Virtual Hosting in the pre-processing stage?** Determining the specific domain targeted if the server services multiple domains.
14. **What does Address Mapping determine?** Whether the request is for static or dynamic content.
15. **What happens during Authentication pre-processing?** Authorization credentials are examined if the resource is protected.
16. **Which module invokes sub-modules to serve content?** The Request Processing module.
17. **What is the function of the Response Generation module?** It builds the response and directs it to the Networking module.
18. **Is a Web server inherently stateful or stateless?** It is stateless.
19. **Does the server itself maintain session information?** No, session

information is maintained by server-side applications.

20. **What must a browser do if a persistent connection is broken?** Re-establish the connection either directly or via a proxy.
21. **What are persistent connections designed for?** To improve performance and continued usage.
22. **What are servers responsible for in the context of persistent connections?** Maintaining queues of requests and responses.
23. **What order must responses be sent back in a persistent connection?** In the order of request arrival (FIFO - First In, First Out).
24. **How does a server manage the order of responses?** By using input and output queues.
25. **When is a request removed from the input queue?** When it is submitted for processing.
26. **What happens to a request on the output queue once processing is complete?** It is marked for release but remains until predecessors are gone.
27. **Under what condition is a response finally sent from the output queue?** When all of its predecessors have been released from the queue.
28. **What does the server do after picking a request from the queue?** Resolves the URL to a physical file location and checks for authentication.
29. **What occurs if authentication fails at this stage?** The server aborts processing and generates an error response.
30. **What are the two categories of static content?** Static content pages and as-is pages.
31. **What defines a "static content page"?** A file (HTML, XML, text, etc.) for which the server must construct a response, including headers.
32. **What defines an "as-is page"?** A page for which complete HTTP responses, including headers, already exist.
33. **What requires "explicit programmatic action" from the server?** Dynamic content.
34. **How do servers determine which processing mechanism to use for a URL?** By using a combination of filename suffixes/extensions and URL prefixes.
35. **What is the default assumption for processing a URL?** That it is a request for a static content page.
36. **What might a URL path starting with /servlet/ indicate?** The target is a Java servlet.

CSE 304 – Web Application Development

Summary

37. **What might a URL path starting with /cgi-bin/ indicate?** The target is a CGI script.

38. **Which file extension often indicates a CGI script?** .cgi.

39. **What extensions indicate template processing like PHP or Cold Fusion?** .php or .cfm.

40. **To what does the server map a static page URL?** A file location relative to the server document root.

41. **If the document root is /www/doc, what is the file path for /pages/school.html?** /www/doc/pages/school.html.

42. **What does the first line of a response (Example 4) contain?** The status code summarizing the operation.

43. **How does the server control browser rendering for static files?** Through the Content-Type header set to a MIME type.

44. **How are MIME types assigned to static files?** Through server configuration, usually based on file extensions .

45. **What is the benefit of "as-is" pages?** The server sends them back unchanged without adding its own headers or status codes.

46. **How can you control server output using "as-is" files?** By manually creating and modifying response messages in a file.

47. **What is a major limitation of "as-is" pages?** They do not provide the opportunity to implement processing logic.

48. **When did CGI and SSI originate?** At the very beginnings of the World Wide Web.

49. **What was the first consistent server-independent mechanism for dynamic content?** CGI.

50. **What does the CGI mechanism assume happens when a request arrives?** A new "process" is spawned/created to execute an application program.

51. **What is the "heart" of the CGI specification?** A fixed set of environment variables accessible to all CGI applications.

52. **Which three environment variables must the server always set?** SERVER_SOFTWARE, SERVER_NAME, and GATEWAY_INTERFACE.

53. **How are CONTENT_TYPE and CONTENT_LENGTH variables populated?** From the corresponding HTTP headers.

54. **How is an HTTP header like User-Agent mapped to a CGI variable?** Upper case letters, replace dash with underscore, prepend HTTP_ (e.g., HTTP_USER_AGENT).

55. **What does the SERVER_PROTOCOL variable represent?** The HTTP version as defined on the request line.

56. **What is stored in the REQUEST_METHOD variable?** The HTTP method used (e.g., GET, POST).

57. **What is PATH_INFO?** Extra path information following the CGI script location in the URL.

58. **What is stored in SCRIPT_NAME?** The path portion of the URL excluding extra path information.

59. **What is QUERY_STRING?** The information following the "?" in a URL.

60. **Do CGI environment variable names have meaning outside the CGI context?** No.

61. **What was the goal of defining CGI as a set of rules?** To ensure programs run the same way on different servers and operating systems.

62. **How does a CGI program receive the body of a request in UNIX?** Through the standard input stream.

63. **What is the lifetime of a CGI process?** A single request.

64. **What is a performance drawback of CGI?** Common processing tasks must be repeated every time because the process terminates after each request.

65. **Is persistent storage a guaranteed feature of CGI?** No, it is a non-standard additional service out of scope of HTTP server guarantees.

66. **What is URL-encoding?** A process where spaces become "+" and punctuation becomes a "%" followed by two hex digits.

67. **What Content-Type is set when a browser submits form data?** application/x-www-form-urlencoded.

68. **Should Content-Type of the request be confused with Content-Type of the response?** No.

69. **What are two ways a server decides a file is a CGI program?** Configuration parameters (CGI directory) or the .cgi extension.

70. **What must a server verify before running a CGI script?** That the location is legal and the script has execute permissions .

71. **What does the response processor add to a CGI response if missing?** Default status code, default Content-Type, and server identification headers.

72. **What are SSI directives?** Directives placed in HTML pages and evaluated on the server while the page is being served.

CSE 304 – Web Application Development

Summary

73. **What is the advantage of SSI over CGI?** It lets you add dynamic content to a page without serving the entire page via a program.
74. **Give an example of a simple SSI directive.** ``.
75. **What is a "hit counter" typically implemented with?** SSI's include virtual function calling a CGI program .
76. **How do you enable SSI in an Apache configuration?** By adding Options +Includes .
77. **Why should you apply SSI Options to specific directories?** To ensure the directive gets evaluated last and isn't overridden.
78. **What is one way to tell Apache to parse files for SSI?** Assigning a specific extension like .shtml.
79. **What is the disadvantage of using the .shtml extension?** You must change the file name and all links to that page.
80. **What does the XBitHack directive do?** It tells Apache to parse files for SSI if they have the execute bit set.
81. **How do you set the execute bit on a file in UNIX?** Using the command chmod +x pagename.html.
82. **Why is it a bad idea to parse all .html files for SSI?** It requires the server to read through every file, which slows down performance .
83. **Why does Apache not send last-modified or content-length for SSI pages by default?** Because these are difficult to calculate for dynamic content.
84. **What is "XBitHack Full" configuration used for?** To tell Apache to use the modification date of the original file for the header.
85. **What is the syntax for an SSI directive?** `` .
86. **Why is the SSI syntax formatted like an HTML comment?** So the browser ignores it if SSI is not enabled.
87. **What does the echo function do in SSI?** It prints out the value of a variable.
88. **Can you define custom variables in SSI?** Yes, using the set function.
89. **How do you change the date format in SSI?** Using the config function with the timefmt attribute.
90. **Which SSI function shows the modification date of a file?** flastmod.
91. **What is the purpose of ``?** To include the output of a CGI script in the page.
92. **How does timefmt="%A %B %d, %Y" display a date?** Full weekday, full month name, day, and year.

93. **What is the LAST_MODIFIED variable?** A generic variable showing the file's last modification time.

94. **What search term provides details on the timefmt format?** strftime.

95. **What is a common use for the include command?** Reducing the burden of updating standard headers and footers .

96. **What is the difference between the file and virtual attributes in an include?** file is relative to the current directory; virtual is a URL relative to the document .

97. **Can the file attribute contain ../?** No.

98. **Can the virtual attribute start with a /?** Yes, but the file must be on the same server.

99. **Can SSI directives be nested?** Yes, an included file can include another file.

100. **Where is a good place to put a LAST_MODIFIED directive?** Inside an included footer file.

UNIT 7

1. **What makes upgrading fielded systems challenging in the current landscape?** A constantly changing technology landscape and expectations to adapt quickly.

2. **What are the drawbacks of highly integrated systems?** They are often proprietary, vendor-locked, expensive, and difficult to upgrade.

3. **What is the primary strategy for overcoming vendor lock-in?** Designing highly interoperable systems.

4. **How is OSA defined?** An integrated business and technical approach to acquire interoperable components using modular design.

5. **What is the business goal of OSA?** To drive down costs, adapt to changing needs, and increase vendor competition.

6. **How does the technical approach of OSA work?** It decomposes systems into components that interact through key interfaces according to formal specifications.

7. **What is interoperability?** The ability of computer systems or software to exchange and make use of information.

8. **What characterizes a system with complete interoperability?** Its interfaces are completely understood, and it works with other systems without restrictions.

9. **Does interoperability include the ability to execute the same binary code on different processors?** No, that is not contemplated by the definition.

10. **What is a common consequence of lacking interoperability?** A lack of attention to standardization during program design.
11. **Name three benefits of OSA for vendor selection.** Increased flexibility, simplified maintenance, and greater access to innovation.
12. **How does OSA affect development processes?** It shortens design times and streamlines development.
13. **What financial benefit does OSA provide?** Reduced total cost of ownership.
14. **Does OSA apply only to computers?** No, it also applies to communications, electricity production, weapons, and vehicles.
15. **How do military vehicles utilize OSA?** Through standard electronic platforms and mounting systems to quickly introduce new weapon and sensor capabilities.
16. **What are the three key elements of OSA?** Components, interfaces, and standards.
17. **What dimensions of interoperability do these elements span?** Technical, informational, and organizational.
18. **What are components?** The physical modules of a system with distinct functionality.
19. **How do components interact with the rest of the system?** They operate independently with limited impact on other parts.
20. **What is the benefit of decoupling components?** It makes it possible to interchange units from alternate vendors.
21. **How are interfaces defined?** As the key boundaries between components and how they interact.
22. **On what do the types of interfaces depend?** The physical connections, information, or services exchanged.
23. **What are three requirements for interfaces in OSA?** They should be standardized, configuration managed, and publicly available.
24. **What do standards define?** The specifications for how components interact through interfaces.
25. **What performance requirements are included in standards?** Security, reliability, and maintainability.
26. **Who should manage standards?** Consensus groups.
27. **When should a program consider interoperability in a system's life cycle?** Through the entire life cycle, from acquisition to decommissioning.

28. **List the four critical success factors for implementing OSA.** 1. Firm commitments/governance, 2. Available/economical components, 3. Controlled interfaces, and 4. Mature standards .
29. **What is required for successful enterprise-wide OSA implementation?** Cooperation and dedicated enterprise-wide strategy.
30. **What role do senior managers play in OSA?** Providing political and financial support based on understanding long-term benefits.
31. **What practices help organizations manage OSA system complexity?** Systems engineering and enterprise architecture.
32. **Why must component specifications be formal and public?** To encourage broad commercial support and standardized functionality across vendors.
33. **What does vendor competition incentivize?** Productivity, reliability, and innovation.
34. **How does a program assess a system's openness over time?** By monitoring the number of interfaces, their rate of change, and conformance with standards.
35. **Why should systems use mature industry standards?** To mitigate risks and ensure interfaces meet industry-wide requirements.
36. **What is the OSI model's relation to OSA?** It decomposes communications into functional layers with well-defined protocols.
37. **How does a service-oriented architecture (SOA) align with OSA?** It separates software into loosely coupled pieces that communicate via standard web services.
38. **What are the five ways to achieve software interoperability?** 1. Product testing, 2. Product engineering, 3. Industry/community partnership, 4. Common technology/IP, and 5. Standard implementation.
39. **Why is formal production testing necessary?** Conformance to a standard does not necessarily guarantee interoperability between different products.
40. **What is the purpose of an industry/community partnership?** To define common standards used to allow systems to intercommunicate for a defined purpose.
41. **How can a community make interoperability more achievable?** By sub-profiling an existing standard to reduce implementation options.
42. **What role do third-party libraries play in interoperability?** They

provide common technology that reduces variability between separately developed products.

43. **What is socket programming?** A way of connecting two nodes on a network to communicate.
44. **What defines one endpoint of a socket?** An IP address and a port number.
45. **What is a socket bound to?** A port number.
46. **Why is a socket bound to a port number?** So the TCP layer can identify the destination application.
47. **What uniquely identifies every TCP connection?** Its two endpoints.
48. **What are the two types of sockets?** Datagram socket and stream socket.
49. **Describe a datagram socket.** A connectionless point for sending and receiving packets, similar to a mailbox.
50. **Describe a stream socket.** A connection-oriented, sequenced flow of data, similar to a phone conversation.
51. **What does the socket() system call do?** Creates a socket.
52. **What does the bind() system call do?** Identifies the socket (like a telephone number).
53. **What does the listen() system call do?** Prepares the socket to receive a connection.
54. **What does the accept() system call do?** Confirms/accepts a connection from a sender.
55. **What does the connect() system call do?** Prepares a client to act as a sender to a server.
56. **What system calls are used to send and receive data?** write() and read() (or send() and recv()).
57. **Which Java package provides socket implementation?** java.net.
58. **What is the purpose of the Java Socket class?** It implements one side of a two-way connection and hides system-dependent details.
59. **What class does a Java server use to listen for clients?** ServerSocket.
60. **When are the URL and URLConnection classes more appropriate than sockets?** When trying to connect to the Web.
61. **Where does a server typically run?** On a specific computer with a socket bound to a specific port.
62. **What must a client know to request a connection?** The server's hostname and the port number it is listening on.
63. **What happens on the server side upon accepting a connection?** The server gets a new socket bound to the same local port but with the

remote endpoint set to the client's address.

64. **Why does the server need a new socket for an established connection?** To continue listening for other connection requests on the original socket.
65. **Which process typically makes the initial request for information?** The client process.
66. **Give an example of a client application.** An Internet Browser.
67. **Give an example of a server process.** A Web Server.
68. **Must a server know a client's address before a connection is established?** No, but the client must know the server's address.
69. **What are the two types of client-server architectures?** 2-tier and 3-tier.
70. **Describe a 2-tier architecture.** The client directly interacts with the server.
71. **What is a risk of 2-tier architecture?** Security holes and performance problems.
72. **How are security problems resolved in a 2-tier web architecture?** Using Secure Socket Layer (SSL).
73. **What is "middleware"?** Software that sits between the client and server in a 3-tier architecture.
74. **What are two functions of middleware?** Security checks and load balancing.
75. **Name two examples of middleware software.** Web Logic or WebSphere.
76. **What is an "Iterative Server"?** A server that serves one client at a time; others must wait.
77. **What is a "Concurrent Server"?** A server that runs multiple processes to serve many requests at once.
78. **How does a Unix server typically handle concurrent requests?** By forking a child process for each client.
79. **What call typically blocks a server until a client connects?** accept().
80. **What is the "TCP three-way handshake"?** The process of connection establishment between client and server.
81. **What is the generic socket address structure in Unix?** sockaddr.
82. **What attribute in sockaddr represents the address family?** sa_family.
83. **Which value is most common for Internet-based address families?** AF_INET.
84. **What structure is specifically used for the Internet family?** sockaddr_in.

85. **What is the size of the IP address in the sockaddr_in structure?** 32-bit.

86. **What is the size of the port number in the sockaddr_in structure?** 16-bit.

87. **What structure holds the 32-bit netid/hostid?** in_addr.

88. **What information does the hostent structure keep?** Information related to the host, like its official name and aliases.

89. **What structure stores information about services and ports?** servent.

90. **How are socket address structures always passed to functions?** By reference (pointers).

91. **What are "value-result" arguments in socket programming?** Arguments where the size is passed by reference and updated by the function.

92. **Why should structure variables be set to NULL (e.g., using memset)?** To avoid unexpected junk values.

93. **What is the range for "well-known" port numbers?** Smaller than 1024.

94. **What is the range for a general port?** Between 1024 and 65535.

95. **What port does HTTP use?** 80.

96. **What port does Telnet use?** 23.

97. **What port does SMTP (email) use?** 25.

98. **What port does FTP use?** 21.

99. **Which function fetches a service name from the /etc/services file using a port number?** getservbyport().

100. **Which function returns a port number when given a service name and protocol?** getservbyname().